

FULL NAME: _____

BLOCK: _____

Multiply & Divide Rational Expressions + DOK-3 Closure

Lesson 4-3 · Period 2 · Student Packet

OBJECTIVES

Math Objective	Multiply and divide rational expressions by factoring, canceling common factors, and rewriting division as multiplication by the reciprocal; track all domain restrictions through every step.
Language Objective	Justify in writing whether a set is closed under an operation, using sentence frames that cite factoring and cancellation.
Essential Understanding	Rational expressions behave like rational numbers: closed under multiplication and nonzero division, because factoring + cancellation always produces another rational expression.

TOPIC GOALS / ESSENTIAL QUESTION / MATERIALS

Topic Goals	Multiply and divide rational expressions; justify closure under these operations.
Essential Question	Why are rational expressions closed under multiplication and division, and what must we track to prove it?
Materials	Student packet, pencil. Teacher models Example 3; DOK-3 Practice #13 is the capstone closure-argument task.

LAUNCH — Multiply & Divide Rational Expressions (teacher models Example 3)

[BOOK] MULTIPLY / DIVIDE RULES (keep printed on your page)

MULTIPLY / DIVIDE RULES

1. Factor every numerator AND denominator completely.
2. For division, rewrite as multiplication by the RECIPROCAL of the divisor.
3. Cancel only common FACTORS across any numerator with any denominator.
4. State the domain: exclude every zero of every denominator that appeared in ANY step, including the reciprocal's new denominator.

[WARN] DIVISION IS MULTIPLICATION BY THE RECIPROCAL

$$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c}$$

The new denominator (c) must be added to your domain-restriction list even if it cancelled.
Never cancel across the $\div$ sign — flip FIRST, then cancel.

EXPLORE — Think-Pair-Share (33 min)

Work with a partner. Move in order. Practice #13 (Closure Argument) is the big one — plan ~12 min for it.

Bank-item labels (in order):

1. Try It 3 — Multiply rational expressions
2. Practice #27 — Multiply with factoring
3. Try It 4 — Divide rational expressions
4. Practice #29 — Divide: rewrite as multiplication
5. Try It 5 — Mixed multiply/divide
6. Practice #32 — Multi-step multiply/divide
7. Practice #13 [STAR] DOK-3 CLOSURE ARGUMENT

Try It 3 — Multiply rational expressions

Find the simplified form of each product, and state the domain.

a. $\frac{x^2 - 16}{9 - x} \cdot \frac{x^2 + x - 90}{x^2 + 14x + 40}$

b. $\frac{x + 3}{4x} \cdot \frac{3x - 18}{6x + 18} \cdot \frac{x^2}{4x + 12}$

Practice #27 — Multiply with factoring

Find the product and the domain.

$$\frac{(x - y)^2}{x + y} \cdot \frac{3x + 3y}{x^2 - y^2}$$

Try It 4 — Divide rational expressions

Find the simplified form of each product and the domain.

a. $\frac{x^3 - 4x}{6x^2 - 13x - 5} \cdot (2x^3 - 3x^2 - 5x)$

b. $\frac{3x^2 + 6x}{x^2 - 49} \cdot (x^2 + 9x + 14)$

Practice #29 — Divide: rewrite as multiplication

Find the product and the domain.

$$\frac{2x^2 - 10x}{(x - 5)(x^2 - 1)} \cdot (3x^2 + 4x + 1)$$

Try It 5 — Mixed multiply/divide

Find the simplified quotient and the domain of each expression.

a. $\frac{1}{x^2 + 9x} \div \left(\frac{6 - x}{3x^2 - 18x} \right)$

b. $\frac{2x^2 - 12x}{x + 5} \div \left(\frac{x - 6}{x + 5}\right)$

Practice #32 — Multi-step multiply/divide

Find the quotient and the domain.

$$\frac{25x^2 - 4}{x^2 - 9} \div \frac{5x - 2}{x + 3}$$

Practice #13 [STAR] DOK-3 CLOSURE ARGUMENT

Higher Order Thinking Why are rational expressions closed under multiplication and division?

SHARE / SUMMARY (5 min)

Sentence frames

Pair share (Closure Argument): “I rewrite the division as multiplication by the reciprocal, giving _____. After factoring, I cancel the factor _____, leaving _____. The domain excludes $x = ___$ because those values make at least one denominator zero at some step.”

Whole class: one pair reads the closure argument. Class signals thumbs up/sideways.

[SEARCH] BRIDGE TO P3

Next period we apply multiply/divide skills to complex rational expressions and assessment-critical items. Bring your domain-restriction list habit.

EXIT

To divide rational expressions I first _____.

I cancel common factors, not _____.

My domain restriction list must include zeros from _____.

One biggest thing I learned today: _____

OPTIONAL REINFORCEMENT (not collected)

If you finish Explore early. #40 is a finance performance task that applies rational expressions to a real-world context.

Optional #1 (Savvas Practice, performance-task finance item)

Performance Task The approximate annual interest rate r of a monthly installment loan is given by the formula:

$$r = \frac{\frac{24(nm-p)}{n}}{p + \frac{nm}{12}}$$

where n is the total number of payments, m is the monthly payment, and p is the amount financed.

Part A Find the approximate annual interest rate (to the nearest percent) for a four-year signature loan of \$20,000 that has monthly payments of \$500.

Part B Find the approximate annual interest rate (to the nearest tenth percent) for a five-year auto loan of \$40,000 that has monthly payments of \$750.